Recording design contract — PROVISIONAL

Purpose

Explain the exact normalized conditional law for a missing block under a supplied joint Gaussian. This separate proof supplements the prepared live residual argument.

Endpoint

Prove the normalized equality and identify its factors by integration and a positive-density ratio.

Entry / assumed context

Live block notation, Gaussian density and centered-residual interpretation; prior matrix algebra, positive definiteness, determinants and integration. The live independence proof is never a premise here.

Narrative arc

P0 introduces the recording; R1 proves valid elimination and quadratic separation; R2 completes normalization and identifies the probability laws. Twenty slides in the supplied order.

Visual and notation anchors

Use $H = \Sigma_{22}$, $B = \Sigma_{12}H^{-1}$, $S = \Sigma_{11} - BHB^T$, $u = x_2 - \mu_2$, $r = x_1 - \mu_1 - Bu$. Matrices retain their exact order.

Summary design brief

Each section closes with its established objects and retrospective takeaways. R2.8 closes the recording.

Continuity / entry context

Inspected live handout 48: dominant conditional formula with adjacent mean proof, intrinsic reasons and separate lower key points. Handout 52: conditional/residual objects retain explicit centering maps. Carry forward this hierarchy; new algebra receives normal proof emphasis, recalled items stay subordinate.

Constraints / open pressure

All slide titles/count/order are provisional upstream commitments preserved here. No timing claim. No new dataset, notebook, live-deck edit or alternate proof. Final acceptance and rehearsal remain instructor/coordinator matters. Source: Murphy Book 2 (2023), online Dec. 10, 2025, §2.3.1.5, printed pp. 25–27; elimination, SPD and explicit normalization completion are course constructions. See math_derivation.md for exact provenance.

Quadratic separation of a Gaussian

Teaching move

Locate this separate density proof after the prepared live Gaussian-conditioning argument.

Layout sketch

subtitle

task

Content objects

subtitle card

A complete density proof of Gaussian conditioning.

task — Key points key-points

Under a supplied joint Gaussian, why is the missing-block conditional exactly normalized?

Reveal sketch

Static: subtitle, task

Design constraints

A recording title, without a new result or practical-competence claim.

R1 — Block elimination separates the joint quadratic

Purpose

Derive a reversible covariance elimination and use its inverse congruence to separate the quadratic.

Endpoint

$$(x-\mu)^T\Sigma^{-1}(x-\mu) = r^T S^{-1} r + u^T H^{-1} u$$

. Every inverse has been justified; normalized factorization is still unproved.

Entry / assumed context

The supplied joint covariance is symmetric positive definite. The Gaussian density and live residual interpretation are available; the desired conditional is not a proof premise.

Narrative arc

Density obligations → observed-block positivity → reversible elimination → two block products → residual positivity → inverse order → residual coordinates → split quadratic.

Visual and notation anchors

$T = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix}$; $D = T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}$. Identity/zero blocks match d_1, d_2. Transposes and inverse order remain explicit.

Summary design brief

Select congruence, block positivity and the quadratic split. Omit the intermediate inverse display from the recap because it already received its own derivation.

Continuity / entry context

Recall cards cite actual live or earlier recording objects. Block cancellations and SPD/inverse arguments are newly taught proof objects. Local labels identify operations; interpretation occupies a separate key-point slot.

Constraints / open pressure

R1.3 and R1.10 are relatively dense; maintain readable normal mathematics and all warrants. Authoring may rebalance subordinate context, but must return any requested title/count/order change.

Block elimination separates the joint quadratic

Teaching move

Introduce the section purpose through its title and subtitle.

Layout sketch

subtitle

Content objects

subtitle card

Proving the algebra behind Gaussian conditioning

Reveal sketch

Static: subtitle

Rationale

Production backpressure resolved provisionally: The former opening question and key point repeated the section objective and introduced no mathematical object. The exact title/subtitle preserves the section opening within its existing slide.

Design constraints

Render this unit as the course section divider. The subtitle object is section metadata, not an extra content card or additional slide. Full title as above; subtitle: Proving the algebra behind Gaussian conditioning; abbreviated footer: Quadratic separation. No theorem answer or new argument is introduced here.

The joint density sets two proof obligations

Teaching move

Recall the complete joint density and distinguish exponent from normalizer.

Layout sketch

model

density

kp

Content objects

model — Supplied block Gaussian recall-card

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}, \quad \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} = \Sigma^T \succ 0$$

$$X_i, \mu_i \in \mathbb{R}^{d_i}, \quad \Sigma_{ij} \in \mathbb{R}^{d_i \times d_j}, \quad d = d_1 + d_2, \quad d_i \geq 1$$

From: Prepared live: Gaussian blocks and density

density card

$$p(x_1, x_2) = \mathcal{N}_d(x; \mu, \Sigma)$$

$$= \underbrace{(2\pi)^{-\frac{d}{2}} (\det \Sigma)^{-\frac{1}{2}}}_{\text{normalizer}} \exp\left(-\frac{1}{2} \underbrace{(x-\mu)^T \Sigma^{-1} (x-\mu)}_{\text{quadratic}} \right)$$

kp — Key points key-points

Separating the quadratic is one obligation. The normalizer must split as well.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	model	—	Initial meaningful context
STATE 2	+ density, kp	—	Inspect the two load-bearing parts of the same density.

Design constraints

Use covariance-parameterized Gaussian notation throughout; do not introduce target conditional as an established identity.

Positive definiteness makes the observed block invertible

Teaching move

Prove observed-block positivity, then define the elimination coefficients.

Layout sketch

positive

definitions

identities

kp

Content objects

positive — Nonzero observed-block test card

$$H := \Sigma_{22}, \quad z \in \mathbb{R}^{d_2} \setminus \{0\}$$

$$z^T H z = \begin{pmatrix} 0 \\ z \end{pmatrix}^T \Sigma \begin{pmatrix} 0 \\ z \end{pmatrix} > 0 \quad \Rightarrow \quad H = H^T \succ 0$$

definitions — Now the inverse exists card

$$B := \Sigma_{12} H^{-1} \in \mathbb{R}^{d_1 \times d_2}, \quad S := \Sigma_{11} - BHB^T = S^T$$

identities card

$$BH = \Sigma_{12}, \quad HB^T = \Sigma_{21}$$

$$S = \Sigma_{11} - \Sigma_{12} H^{-1} \Sigma_{21}$$

kp — Key points key-points

S is the live residual covariance (the Schur complement of H). Its strict positivity still needs proof here.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	positive	—	Initial meaningful context
STATE 2	+ definitions	entry emphasis	Define B and S only after proving H invertible.
STATE 3	+ identities, kp	—	Use symmetry of H inverse and Sigma21=Sigma12 transpose.

Design constraints

Intrinsic implication and symmetry warrant stay with the proof. Do not display S inverse yet.

The elimination matrix has an explicit inverse

Teaching move

Verify reversibility by multiplication rather than importing a block-inverse lemma.

Layout sketch

matrices

products

kp

Content objects

matrices card

$$T := \begin{pmatrix} I_{d_1} & -B \\ 0 & I_{d_2} \end{pmatrix}, \quad U := \begin{pmatrix} I_{d_1} & B \\ 0 & I_{d_2} \end{pmatrix}$$

products — Multiply in both orders card

$$TU = \begin{pmatrix} I_{d_1} & B - B \\ 0 & I_{d_2} \end{pmatrix} = I_d$$

$$UT = \begin{pmatrix} I_{d_1} & -B + B \\ 0 & I_{d_2} \end{pmatrix} = I_d \quad \Rightarrow \quad T^{-1} = U$$

kp — Key points key-points

Block elimination is reversible; no covariance or independence argument was used.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	matrices	—	Initial meaningful context
STATE 2	+ products, kp	entry emphasis	Check the off-diagonal cancellation in both orders.

Design constraints

After this slide I and 0 may omit subscripts when block dimensions are fixed.

Left elimination removes the upper covariance block

Teaching move

Work the first block multiplication, including its upper-right cancellation.

Layout sketch

recall

multiplication

result

kp

Content objects

recall — Carried forward recall-card

$$BH = \Sigma_{12}, \quad HB^T = \Sigma_{21}, \quad S = \Sigma_{11} - BHB^T$$

From: R1.3

multiplication — Left multiplication card

$$T\Sigma = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix} \begin{pmatrix} \Sigma_{11} & BH \\ HB^T & H \end{pmatrix}$$

$$= \begin{pmatrix} \Sigma_{11} - BHB^T & BH - BH \\ HB^T & H \end{pmatrix}$$

result card

$$T\Sigma = \begin{pmatrix} S & 0 \\ HB^T & H \end{pmatrix}$$

kp — Key points key-points

The upper cross block vanishes. This single multiplication is not yet the covariance congruence.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall	—	Initial meaningful context
STATE 2	+ multiplication	entry emphasis	Show all four product blocks; newly calculated cancellation gets entry emphasis.
STATE 3	+ result, kp	—	Identify S without suppressing the remaining lower cross block.

Design constraints

Final complete state preserves multiplication and result; no independence language.

Right elimination completes the covariance congruence

Teaching move

Carry the preceding product forward and cancel its lower cross block.

Layout sketch

recall

product

diagonal

kp

Content objects

recall — Carried forward recall-card

$$T\Sigma = \begin{pmatrix} S & 0 \\ HB^T & H \end{pmatrix}, \quad T^T = \begin{pmatrix} I & 0 \\ -B^T & I \end{pmatrix}$$

From: R1.5 and transpose of R1.4

product — Right multiplication card

$$T\Sigma T^T = \begin{pmatrix} S & 0 \\ HB^T & H \end{pmatrix} \begin{pmatrix} I & 0 \\ -B^T & I \end{pmatrix}$$

$$= \begin{pmatrix} S & 0 \\ HB^T - HB^T & H \end{pmatrix}$$

diagonal card

$$D := T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}$$

kp — Key points key-points

The two multiplications form a congruence: $T\Sigma T^T$. They do not form a similarity transformation.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall	—	Initial meaningful context
STATE 2	+ product	entry emphasis	Students can compare the inherited lower block with its cancellation.
STATE 3	+ diagonal, kp	—	Name the diagonal covariance object D.

Summary candidates

VISUAL diagonal — Central covariance anchor for the retrospective.

Invertible congruence proves the residual covariance positive definite

Teaching move

Prove D positive definite, then restrict its quadratic to the first block.

Layout sketch

recall

congruence

restriction

kp

Content objects

recall — Carried forward recall-card

$$D = T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}, \quad T \text{ invertible}, \quad \Sigma \succ 0$$

From: R1.4 and R1.6

congruence — Every nonzero test vector card

$$w \neq 0 \quad \Rightarrow \quad T^T w \neq 0$$

$$w^T D w = (T^T w)^T \Sigma (T^T w) > 0 \quad \Rightarrow \quad D \succ 0$$

restriction — Test the first block card

$$a \in \mathbb{R}^{d_1} \setminus \{0\}, \quad w = \begin{pmatrix} a \\ 0 \end{pmatrix}$$

$$a^T S a = w^T D w > 0 \quad \Rightarrow \quad S \succ 0$$

kp — Key points key-points

Both S and H have inverses and positive determinants. This follows from strict joint positivity.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall	—	Initial meaningful context
STATE 2	+ congruence	entry emphasis	Invertibility prevents a nonzero vector from becoming zero.
STATE 3	+ restriction, kp	entry emphasis	The first diagonal block inherits strict positivity.

Summary candidates

VISUAL restriction — Explains why the residual inverse and normalizer are valid.

Inverting the congruence fixes the order of the factors

Teaching move

Derive the inverse identity in its exact matrix order.

Layout sketch

recall

invert

isolate

kp

Content objects

recall — Carried forward recall-card

$$D = T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}$$

From: R1.6; inverses justified in R1.7

invert — Invert the product card

$$D^{-1} = T^{-T}\Sigma^{-1}T^{-1}, \quad T^{-T} := \left(T^{-1}\right)^T$$

isolate — Multiply on the left by T transpose and on the right by T card

$$T^T D^{-1} T = \Sigma^{-1}$$

$$\Sigma^{-1} = T^T \begin{pmatrix} S^{-1} & 0 \\ 0 & H^{-1} \end{pmatrix} T$$

kp — Key points key-points

The factors do not commute. The order follows from undoing the congruence.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall	—	Initial meaningful context
STATE 2	+ invert	entry emphasis	Reverse the factor order when inverting.
STATE 3	+ isolate, kp	entry emphasis	Cancel only adjacent inverse pairs.

Design constraints

Preserve covariance congruence alongside inverse proof. Expanded block inverse is checking material only.

Elimination turns the centered vector into a residual and an observation

Teaching move

Work the coordinate transformation and identify both blocks.

Layout sketch

coordinates

transform

kp

Content objects

coordinates — Centered coordinates card

$$u := x_2 - \mu_2, \quad r := x_1 - \mu_1 - Bu$$

transform card

$$T(x-\mu) = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix} \begin{pmatrix} x_1 - \mu_1 \\ u \end{pmatrix}$$

$$= \begin{pmatrix} x_1 - \mu_1 - Bu \\ u \end{pmatrix} = \begin{pmatrix} r \\ u \end{pmatrix}$$

kp — Key points key-points

r is the centered residual. Its coordinates contain x_2; the original first block is still x_1.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	coordinates	—	Initial meaningful context
STATE 2	+ transform, kp	entry emphasis	Elimination subtracts the predicted deviation while retaining the observation.

Design constraints

No independence statement is needed for this coordinate identity.

The joint quadratic is the sum of two block quadratics

Teaching move

Substitute the proved inverse and transformed vector into the quadratic.

Layout sketch

recall

substitute

split

kp

Content objects

recall — Carried forward recall-card

$$\Sigma^{-1} = T^T D^{-1} T, \quad T(x-\mu) = \begin{pmatrix} r \\ u \end{pmatrix}$$

$$D^{-1} = \begin{pmatrix} S^{-1} & 0 \\ 0 & H^{-1} \end{pmatrix}$$

From: R1.8 and R1.9

substitute — Substitute without changing factor order card

$$(x-\mu)^T \Sigma^{-1} (x-\mu) = [T(x-\mu)]^T D^{-1} [T(x-\mu)]$$

$$= \begin{pmatrix} r \\ u \end{pmatrix}^T \begin{pmatrix} S^{-1} & 0 \\ 0 & H^{-1} \end{pmatrix} \begin{pmatrix} r \\ u \end{pmatrix}$$

split card

$$(x-\mu)^T \Sigma^{-1} (x-\mu) = r^T S^{-1} r + u^T H^{-1} u$$

kp — Key points key-points

The exponent separates exactly. A normalized density equality has not yet been proved.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall	—	Initial meaningful context
STATE 2	+ substitute	entry emphasis	Read the substitution as one whole equality, then the block product.
STATE 3	+ split, kp	entry emphasis	Diagonal inverse gives the sum of two quadratic forms.

Summary candidates

VISUAL split — Section endpoint: split quadratic in residual/observed coordinates.

Design constraints

The inverse notation is algebra, not a recommendation to form numerical inverses. Keep the first equality syntactically whole.

Retrospective: elimination separates the quadratic

Teaching move

Retell the established algebraic argument, ending the section retrospectively.

Layout sketch

summary

kp

Content objects

summary — Summary visual section-summary-visual

$$T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}, \quad H \succ 0, \quad S \succ 0$$

$$(x-\mu)^T \Sigma^{-1} (x-\mu) = r^T S^{-1} r + u^T H^{-1} u$$

$$u = x_2 - \mu_2, \quad r = x_1 - \mu_1 - Bu$$

kp — Summary takeaways section-summary-takeaways

Reversible elimination yields valid residual and observed covariance blocks. The inverse congruence separates their quadratic contributions.

Reveal sketch

Static: summary, kp

Rationale

Select congruence + positivity + final quadratic; the intermediate inverse formula is deliberately omitted to keep the recap recognizable and readable.

Design constraints

Retrospective end: no next-section teaser, question or handoff.

R2 — Normalizers identify the conditional and marginal densities

Purpose

Complete the normalized density equality and identify the factors without importing the target law.

Endpoint

$$p(x_1, x_2) = p(x_1|x_2)p(x_2)$$

with the two Gaussian laws established by integration and division.

Entry / assumed context

R1 proved the quadratic split and $H, S \succ 0$. The determinant and dimension factors in the joint density remain to be separated.

Narrative arc

Unit-determinant elimination → determinant split → dimension split → full Gaussian factors → integrate → divide → interpret dependence.

Visual and notation anchors

Retain r, u, S, H, B and $D = T\Sigma T^T$. Introduce $g_1(x_1; x_2)$ and $g_2(x_2)$ as normalized mathematical factors before assigning probability-law names.

Summary design brief

Select normalized product, its parameter labels and the integration/ratio identification. The takeaway recalls that quadratic, determinant and dimension factors all split; do not repeat every algebraic display.

Continuity / entry context

R2.1 receives R1's quadratic split. Normalizers remain next to their matching exponent/dimension. Prose explaining what the factorization establishes stays in the key-point slot.

Constraints / open pressure

R2.4 full factors and R2.5 integral are load-bearing. Do not replace equality with proportionality or remove normalizers to fit. Production may choose stack versus adjacent factor regions if semantic pairing and readability survive; count/order repair returns to Organization.

Normalizers identify the conditional and marginal densities

Teaching move

Receive the established exponent split through the section subtitle and frame normalization as the remaining proof job.

Layout sketch

subtitle

Content objects

subtitle card

From separated exponents to probability laws

Reveal sketch

Static: subtitle

Rationale

Production backpressure resolved provisionally: The former quadratic/SPD recall repeats R1.7, R1.10 and the immediately preceding R1.11 retrospective; no new mathematical claim is omitted. The exact title/subtitle preserves the section opening within its existing slide.

Design constraints

Render this unit as the course section divider. The subtitle object is section metadata, not an extra content card or additional slide. Full title as above; subtitle: From separated exponents to probability laws; abbreviated footer: Normalized factorization. No theorem answer or new argument is introduced here.

Unit-determinant elimination splits the covariance determinant

Teaching move

Use the established congruence and block triangular determinant.

Layout sketch

recall

unit

determinants

kp

Content objects

recall — Carried forward recall-card

$$T = \begin{pmatrix} I & -B \\ 0 & I \end{pmatrix}, \quad D = T\Sigma T^T = \begin{pmatrix} S & 0 \\ 0 & H \end{pmatrix}$$

From: R1.4 and R1.6

unit — Block triangular T card

$$\det T = \det I_{d_1} \det I_{d_2} = 1$$

determinants — Take determinants of the congruence card

$$\det D = (\det T)(\det \Sigma)(\det T^T) = \det \Sigma$$

$$\det \Sigma = \det S \det H, \quad \det \Sigma, \det S, \det H > 0$$

kp — Key points key-points

Elimination changes the covariance blocks while preserving this determinant. Positivity licenses positive square roots.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall	—	Initial meaningful context
STATE 2	+ unit	entry emphasis	The identity diagonal blocks make the determinant one.
STATE 3	+ determinants, kp	entry emphasis	Combine multiplicativity with the block diagonal determinant.

Summary candidates

VISUAL determinants — The determinant obligation is discharged.

The dimensions split the remaining Gaussian normalizer

Teaching move

Keep determinant and dimension factors visible in the same exact identity.

Layout sketch

recall

substitution

split

kp

Content objects

recall — Carried forward recall-card

$$d = d_1 + d_2, \quad \det \Sigma = \det S \det H, \quad S, H \succ 0$$

From: R1.2, R1.7 and R2.2

substitution — Substitute both identities card

$$(2\pi)^{-\frac{d}{2}} (\det \Sigma)^{-\frac{1}{2}}$$

$$= (2\pi)^{-\frac{d_1+d_2}{2}} (\det S \det H)^{-\frac{1}{2}}$$

split — Separate positive factors card

$$(2\pi)^{-\frac{d}{2}} (\det \Sigma)^{-\frac{1}{2}}$$

$$= \left[(2\pi)^{-\frac{d_1}{2}} (\det S)^{-\frac{1}{2}}\right] \left[(2\pi)^{-\frac{d_2}{2}} (\det H)^{-\frac{1}{2}}\right]$$

kp — Key points key-points

Each block receives its own determinant factor and its own dimension factor.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall	—	Initial meaningful context
STATE 2	+ substitution	entry emphasis	Replace d and determinant Sigma before splitting.
STATE 3	+ split, kp	entry emphasis	Pair dimension d1 with S and dimension d2 with H.

Summary candidates

VISUAL split — Normalizers split exactly, including 2 pi powers.

Design constraints

The final equality may occupy two lines at its equals sign. Keep each bracketed block normalizer intact; do not use a proportionality symbol.

The full density equals two normalized Gaussian factors

Teaching move

Associate each full normalizer with its quadratic, then show the exact product.

Layout sketch

recall

factor1

factor2

product

kp

Content objects

recall — Carried forward recall-card

$$u = x_2 - \mu_2, \quad r = x_1 - (\mu_1 + Bu), \quad S, H \succ 0$$

From: R1.3, R1.7 and R1.9

factor1 — First normalized factor card

$$g_1(x_1; x_2) := (2\pi)^{-\frac{d_1}{2}} (\det S)^{-\frac{1}{2}} \exp\left(-\frac{1}{2} r^T S^{-1} r\right)$$

$$= \mathcal{N}_{d_1}(x_1; \mu_1 + Bu, S)$$

factor2 — Second normalized factor card

$$g_2(x_2) := (2\pi)^{-\frac{d_2}{2}} (\det H)^{-\frac{1}{2}} \exp\left(-\frac{1}{2} u^T H^{-1} u\right)$$

$$= \mathcal{N}_{d_2}(x_2; \mu_2, H)$$

product card

$$p(x_1, x_2) = g_1(x_1; x_2) g_2(x_2)$$

kp — Key points key-points

For fixed x_2, g_1 is a Gaussian density in x_1. The factor names do not yet identify the marginal and conditional.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall	—	Initial meaningful context
STATE 2	+ factor1	entry emphasis	Pair the d1 normalizer and residual quadratic.
STATE 3	+ factor2	entry emphasis	Pair the d2 normalizer and observed quadratic; first factor persists.
STATE 4	+ product, kp	—	Combine R1.10 exponent split and R2.3 normalizer split.

Summary candidates

VISUAL product — Exact normalized factorization before probability-law identification.

Design constraints

Keep both full normalized factors, parameter labels and subordinate residual definitions. Never replace equality by proportionality or name the probability laws before R2.5–R2.6.

Integrating the first factor identifies the marginal

Teaching move

Use normalization in x1 for fixed x2, then integrate the joint product.

Layout sketch

recall

unitintegral

marginal

kp

Content objects

recall — Carried forward recall-card

$$p(x_1, x_2) = g_1(x_1; x_2) g_2(x_2), \quad g_1(x_1; x_2) = \mathcal{N}_{d_1}(x_1; \mu_1 + Bu, S)$$

From: R2.4

unitintegral — Fix x2; integrate in x1 card

$$S \succ 0, \quad m \coloneqq \mu_1 + B(x_2 - \mu_2) \ \text{ is fixed}$$

$$\int_{\mathbb{R}^{d_1}} g_1(x_1; x_2)\, \mathrm{d}x_1 = \int_{\mathbb{R}^{d_1}} \mathcal{N}_{d_1}(x_1; m, S)\, \mathrm{d}x_1 = 1$$

marginal — Marginalize card

$$\begin{aligned} p(x_2) &= \int_{\mathbb{R}^{d_1}} p(x_1, x_2)\, \mathrm{d}x_1 \\ &= g_2(x_2) \int_{\mathbb{R}^{d_1}} g_1(x_1; x_2)\, \mathrm{d}x_1 \\ &= g_2(x_2) = \mathcal{N}_{d_2}(x_2; \mu_2, H) \end{aligned}$$

kp — Key points key-points

Normalization identifies g_2 as the marginal; no conditional formula was assumed.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	recall	—	Initial meaningful context
STATE 2	+ unitintegral	entry emphasis	The fixed observation translates the mean but does not change the Gaussian integral.
STATE 3	+ marginal, kp	entry emphasis	The second factor is constant with respect to x1.

Summary candidates

VISUAL marginal — Integration supplies the first probability-law identity.

Design constraints

Gaussian normalization is prior mathematics; no new Jacobian/determinant proof is required. Keep the integration domain, variable, fixed-x2 condition and SPD warrant visible.

Dividing by the positive marginal identifies the conditional

Teaching move

Establish positivity and identify the conditional using the density ratio.

Layout sketch

positive

ratio

identity

kp

Content objects

positive — Every observation has positive marginal density card

$$p(x_2) = g_2(x_2) = \mathcal{N}_{d_2}(x_2; \mu_2, H) > 0 \quad \left(x_2 \in \mathbb{R}^{d_2}\right)$$

ratio — Density ratio card

$$\begin{aligned} p(x_1|x_2) &:= \frac{p(x_1, x_2)}{p(x_2)} \\ &= \frac{g_1(x_1; x_2) g_2(x_2)}{g_2(x_2)} = g_1(x_1; x_2) \\ &= \mathcal{N}_{d_1}(x_1; \mu_1 + B(x_2 - \mu_2), S) \end{aligned}$$

identity card

$$p(x_1, x_2) = p(x_1|x_2)p(x_2)$$

kp — Key points key-points

The positive-density ratio gives a conditional-density version for every x_2. Its normalization was already proved.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	positive	—	Initial meaningful context
STATE 2	+ ratio	entry emphasis	Divide by the marginal identified in R2.5.
STATE 3	+ identity, kp	—	Name the normalized factors only after the integral and ratio.

Summary candidates

VISUAL identity — Final normalized probability-law identity.

Design constraints

Do not interpret a continuous singleton as a positive-probability event or claim uniqueness of conditional versions at null events.

The conditional mean retains the observed-block dependence

Teaching move

Interpret the law and distinguish original blocks from residual coordinates.

Layout sketch

law

parameters

kp

Content objects

law — Carried forward recall-card

$$p(x_1|x_2) = \mathcal{N}_{d_1}(x_1; \mu_1 + B(x_2 - \mu_2), S)$$

From: R2.6

parameters card

$$\mu_{1|2}(x_2) = \mu_1 + B(x_2 - \mu_2), \quad \Sigma_{1|2} = S$$

$$r = x_1 - \mu_{1|2}(x_2)$$

kp — Key points key-points

The mean is affine in x_2 and can depend on the observation; the covariance does not. A separated residual quadratic does not assert that X_1 and X_2 are independent.This exact density proof assumes a joint Gaussian with $\Sigma \succ 0$; it does not establish adequacy of the supplied model.

Reveal sketch

State	Change	Attention	Teaching purpose
OPEN	law	—	Initial meaningful context
STATE 2	+ parameters, kp	—	The x2 dependence remains in the first Gaussian factor through its mean.

Design constraints

No singular-Gaussian extension or new algorithm. The wording includes $B = 0$ and directions in the kernel of B.

Retrospective: normalization completes Gaussian conditioning

Teaching move

Close the recording with the normalized factors and their noncircular identification.

Layout sketch

summary

kp

Content objects

summary — Summary visual section-summary-visual

$$p(x_1, x_2) = g_1(x_1; x_2) g_2(x_2)$$

$$g_1 = \mathcal{N}_{d_1}(x_1; \mu_1 + B(x_2 - \mu_2), S), \quad g_2 = \mathcal{N}_{d_2}(x_2; \mu_2, H)$$

$$\int_{\mathbb{R}^{d_1}} g_1(x_1; x_2)\, \mathrm{d}x_1 = 1 \quad \Rightarrow \quad p(x_2) = g_2(x_2)$$

$$\frac{p(x_1, x_2)}{p(x_2)} = g_1(x_1; x_2) = p(x_1 | x_2)$$

kp — Summary takeaways section-summary-takeaways

The quadratic, determinant and dimension factors all split. Integration then division identify the Gaussian marginal and conditional.

Reveal sketch

Static: summary, kp

Rationale

Select normalized product and identification from R2.4–R2.6. The normalizer identities are recalled in the takeaway, not duplicated as another dense proof. Dependence interpretation already has its own preceding slide.

Design constraints

Final recording slide. Retrospective only; no new claim, task, assignment, handoff or forward question.